

SmartLight Pisa: Comprehensive Development Roadmap v4.2

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Abstract

This document outlines the detailed technical roadmap for the SmartLight Pisa project. It details the transition from theoretical reinforcement learning models to realistic physical simulations using SUMO, and the subsequent scaling to multi-agent coordination and real-world deployment. The current focus is on bridging the simulation-to-reality gap (Phase 3.5).

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1 Introduction & Executive Summary

SmartLight Pisa aims to revolutionize urban traffic management using Deep Reinforcement Learning (DRL). As of February 2026, the project has successfully implemented high-performance DRL agents (PPO, Double DQN) in a simplified environment. The immediate roadmap prioritizes high-fidelity simulation integration to validate these agents against real-world physics before scaling to network-level control.

2 Current Status (v4.0 Completed)

The system currently **successfully implements**:

- **Core Algorithms:** PPO (Proximal Policy Optimization) with Actor-Critic architecture and Dueling Double DQN.
- **High-Performance Training:** Native CUDA GPU acceleration resulting in training speeds of 125 FPS on GTX 1650.
- **State Optimization:** 11-dimensional compressed state space representing queue lengths and wait times. **Parallelism:** Vectorized environments for simultaneous data collection.

3 Phase 3.5: Realistic Simulation Integration (Completed)

[COMPLETED - MARCH 2026]

This phase successfully bridged the gap between mathematical approximation and physical reality.

- **SUMO Integration:** Full environment wrapper completed ('smartlight_sumoenv.py'). **Pisa Digital Twin** *Successfully imports real Pisa road network and traffic flows.*
- **PPO v4.0 Validation:** The Actor-Critic agent achieved an average waiting time of **2.9s/step** on the real map, significantly outperforming fixed-time baselines.

4 Phase 4: Multi-Agent Coordination (Current Focus)

[IN PROGRESS]

Now that single-agent policies are validated, we are scaling to a network of intersections.

- **Grid Simulation:** 3x3 grid of connected intersections in SUMO to test wave-coordination.
- **MAPPO Implementation:** Transitioning from independent agents to Multi-Agent PPO with shared critic for global optimization.
- **V2I/I2I Communication:** Implementing message-passing layers where upstream intersections notify downstream ones about incoming traffic platoons.

5 Phase 5: Real-World Perception & Edge Deployment

[STARTING]

- **Edge AI:** Validation of PPO inference on NVIDIA Jetson Orin Nano hardware.
- **Computer Vision:** Integration of YOLOv8 for vehicle counting, replacing simulator-direct data.
- **Shadow Mode:** Deployment of the agent in a non-controlling "monitoring only" mode for safety certification.

6 Implementation Timeline & Priority Matrix

Priority	Feature	Phase	Est. Complexity	Status
1	Multi-Agent Network Setup (SUMO)	4	High	Next
2	MAPPO Algorithm Development	4	High	Pending
3	I2I Communication Protocol	4	Very High	Pending
4	Jetson Orin Hardware Benchmarking	5	Medium	In Progress
5	YOLOv8 Vision Integration	5	High	Pending

Table 1: Updated Implementation Priority

7 Conclusion

The completion of Phase 3.5 marks the successful validation of SmartLight’s core AI in a high-fidelity environment. The focus now shifts from ”can an AI control an intersection?” to ”can a network of AI agents optimize an entire city?”. This transition to Phase 4 is the final technical hurdle before commercial deployment.